TEAM_CONTEXT — Business Entity Resolution

Amazon ML Challenge 2026 · Team **camelcase** · Handoff document compiled 2026-09-26 from the project's own files: its reports, logs, configs and saved artifacts.

Labels used: MEASURED = a number read from a project artifact (the path is given). HYPOTHESIS = an inference, not proven. **UNRESOLVED** = cannot be determined from the project files.

Scope: Documentation only. Writing it changed no code, model, threshold, config, output or package.

0. Executive summary (one page)

Item	Value
Current public leaderboard score	**0.924**. After uploading `output/matching_results.tsv` on 2026-09-26, the team reported this score in the working session. **It is not recorded in any project file.** The public/private split size and the team's rank are **UNRESOLVED**.
Local validation score (NOT the leaderboard)	Official macro F0.5 = 0.96285 on the held-out 20% validation split (441,364 S1). Source: `experiments/matching/MATCHING_REPORT.md`.
Candidate recall (blocking ceiling)	98.214% on validation. 95.631% when validation S1 are matched against the full training pool (10.3M records). 95.646% for final training (all train S1 × full train pool). Test recall has no labels: **UNRESOLVED**.
Architecture	• Normalisation, then blocking: TF-IDF top-10 per source for a name+address channel and an address channel, both within the query's country, plus 16 exact-key joins with a block cap of 100. • 50 pair features, scored by a numba histogram GBDT written from scratch: 300 trees, depth 6, learning rate 0.1. • No external data and no pretrained models.
Final decision rule	Probability ≥ **0.69**, plus the **one-S1-per-target** rule: each S2/S3 record goes to at most one S1, the one with the highest probability.
Biggest known weaknesses (measured)	• Orphaned-target sensitivity: stress F0.5 is 0.840, driven by false positives on the target-context features. • Native-script names: 89.26% blocking recall. • Empty target addresses: 90.02% blocking recall. • House-number conflicts. • Near-duplicates that differ only in legal form, which cause false merges. • France: unlabelled, **performance UNRESOLVED**.
Current diagnostic status	**COMPLETE** The `t_rank` diagnostic in `experiments/diagnostics/trank/` exited with code 0. Nothing is running now. • Removing, capping or taking the percentile of `t_rank` alone does not change results: `t_gap` takes its place. • Removing all three target-context features helps only in the extreme stress environment. • At the test-like pool ratio (environment T) the current model scores 0.9543.
Exact project state	Steps 1–3, the stress evaluation, final training and inference, the submission file and `camelcase_submission.zip` are all **COMPLETE**. The submission has been uploaded (score as above). No model change has been approved, and the production files are unchanged; hashes were checked at the end of the diagnostic.
Immediate next investigation	The team has **not yet approved** a next step. The candidate directions come from the diagnostic and are documented without ranking in §13: unseen-country robustness, orphan-aware training / calibration, threshold hedging, native-script and empty-address handling, and blocking recall.

Gap accounting, from measured numbers:

- Validation 0.963 → test-ratio environment T 0.954: about −0.009, **measured**.
- T 0.954 → public 0.924: about −0.030, **cause UNRESOLVED**.

- **Hypothesis:** France (15% of test, no labels) is a likely contributor. This is not proven.

1. Project overview

1.1 Problem in simple terms

- There are three independent sources of business records (name, address, country) with no shared identifiers.
- **Source 1 (S1)** is the deduplicated reference list. **S2 and S3** are noisy sources that are *not* internally deduplicated: one business can appear several times in each.
- Task: for **every S1 entity in test**, list all S2/S3 records that are the same real-world business. That can be zero, one or many records (training maximum: 11 per S1).
- Matching goes strictly S1 → {S2, S3}; outputs contain only S2-/S3- IDs. It must be an ML solution trained only on the provided data.

Sources: `docs/problem_statement.pdf`, and `docs/COMPETITION_NOTES.md` §1–§11, which carries page references to the PDF.

1.2 Required output

File	Header	Rules
`output/matching_results.tsv` (the only scored file)	`source1_entity_id<TAB>matched_entity_ids`	• Exactly one row per test S1, with an empty list written as `S1-x<TAB>`. • IDs are comma-separated with no spaces and no duplicates. • Only S2/S3 IDs that exist in test are allowed.
`output/candidate_pairs.tsv` (not scored)	`source1_entity_id<TAB>candidate_entity_ids`	• The exact final candidate set the model scores. • Invariant: matched ⊆ candidates. The validator warns if this is violated.

1.3 Metric: macro F0.5

- Per S1 entity: P = |pred ∩ true| / |pred|, R = |pred ∩ true| / |true|, and **F0.5 = 1.25·P·R / (0.25·P + R)**. The score is the **average over all S1 entities**, singletons included.
- Singletons (no true match): an empty prediction scores **1.0**; **any** prediction scores **0.0**. 5.585% of training S1 are singletons.
- β = 0.5 weights precision about 4× more than recall. One false merge on a singleton wipes out that entity's whole score, which is why the pipeline uses a conservative threshold and the one-per-target rule.
- Two cases are **UNRESOLVED** in the official rules: true ≠ ∅ with an empty prediction, and true ≠ ∅ with no overlap. Both are scored 0.0 locally (R = 0), in `src/ber/metrics.py`; see COMPETITION_NOTES §4.
- Leaderboard: the public score uses a subset of test; the final ranking uses the private remainder. The split size is **UNRESOLVED**.

1.4 Constraints (official materials, via COMPETITION_NOTES §5–§7)

- **External data is strictly prohibited.** That includes external databases, APIs, entity-resolution services, government registries, geocoding, and data augmentation from the internet. Evidence of external lookup means disqualification.
- **Model licence and size:** the final model must be MIT/Apache-2.0 and have at most 8B parameters. Our matcher is a GBDT trained from scratch, and no pretrained weights are used anywhere.
- **Format:** an invalid format is not scored. Missing S1 rows, duplicate rows, duplicate IDs, S1 self-matches and non-existent IDs are rejected.
- **Final zip** (`<team>_submission.zip`):
  - `output/` with both TSVs
  - `code/business_entity_resolution/` with `src/`, `README.md` and a pinned `requirements.txt`; it must be self-contained and runnable

    - the filled `Documentation_template.md`
- Test contains an **unseen country, France** (15% of test S1). The pipeline must treat country as an open set.
- Deadlines, submission rate limits and compute limits: **UNRESOLVED** (not in the materials; COMPETITION_NOTES §12).

1.5 Data sizes (measured, `experiments/EDA_REPORT.md`, COMPETITION_NOTES §13)

File	Rows	Countries
train_source1 / 2 / 3	2,206,821 / 5,034,616 / 5,285,603	US ~60%, India ~40%
train_ground_truth	2,206,821 rows, 7,638,365 true pairs	123,247 singletons (5.58%)
test_source1 / 2 / 3	1,732,544 / 4,887,273 / 5,082,316	India 46.8%, US 38.3%, France 15.0%

Measured structural facts:

- No S2/S3 record matches more than one S1.
- There are no cross-country true pairs.
- About 26% of S2/S3 records match no S1 (distractors).
- Test has 5.75 S2+S3 records per S1, against 4.68 in train. The reason is **UNRESOLVED**.

2. Current project status

Step	Status	What was done	Important result	Main files
EDA	COMPLETE	Streaming EDA over the full data	Noise profile, key recall, singleton rate 5.58%	`experiments/eda.py`, `EDA_REPORT.md`
Step 1: validation + scorer	COMPLETE	Leakage-safe 80/20 split at S1 level; official-metric scorer; tests	441,364 validation S1; the validator passes on the validation ground truth; scorer sanity checks OK	`experiments/validation_split/`, `src/ber/metrics.py`, `src/score.py`
Step 2: blocking	COMPLETE	Design chosen on the fit fold; measured on validation and at full-pool scale	Recall 98.214%, 58.5 candidates/S1 (validation); 95.631% (full pool)	`experiments/blocking/`, `blocking_config.json`
Step 3: matcher	COMPLETE	50 features; GBDT vs logistic regression; selection on the internal A2 split	Validation F0.5 0.96285 (logistic regression 0.94458)	`experiments/matching/`
Test-scale check (unlabelled)	COMPLETE	Step 3 model run on the test set, counts only	t-feature drift on test is negligible (PSI 0.03–0.07)	`experiments/test_scale/`
Labelled stress evaluation	COMPLETE	Validation S1 × full training pool	F0.5 0.840; 0.956 restricted to validation-pool records	`experiments/TEST_SCALE_REPORT.md`, `experiments/test_scale/stress/`
Final training + inference	COMPLETE	Retrained on all training S1 × full pool; test inference	5,503,470 predictions for 1,732,544 S1; validator PASS	`experiments/final/`, `experiments/final_pipeline.py`
Submission + package	COMPLETE	Uploaded `matching_results.tsv`; built and verified the zip	Public LB 0.924 (team-reported; not in files)	`output/`, `submission_package/`, `camelcase_submission.zip`
Full-scale run of the packaged pipeline	PARTIAL	Packaged `run_pipeline.py` only smoke-tested (bit-identical to the smoke reference)	Full-scale reproduction not performed: **UNRESOLVED**	`experiments/final/packaging/`
`t_rank` diagnostic	COMPLETE	Variants A/B/B2/C/D × environments V/S/T	See §9	`experiments/diagnostics/trank/`

Cause of the validation→leaderboard gap	**UNRESOLVED**	—	About 0.009 of the gap is explained by the orphan effect at test ratio; the rest (about 0.03) is not	§9, §12

Nothing is **CURRENTLY RUNNING**. The last background job, the `t_rank` diagnostic, finished with exit code 0.

3. Step 1: validation split and local scorer

3.1 How the split was built

Source: `src/build_validation_split.py` and `src/ber/validation_split.py`; report `experiments/validation_split/VALIDATION_SPLIT.md`.

- **The unit of splitting is the S1 entity**, not the pair: 1/5 of S1 go to validation, with seed `20260925`.
- **Stratification:** country × min(#true matches, 8) × (has an S2 match, has an S3 match), giving 38 non-empty strata. The largest per-stratum deviation from exactly 20% is 0.60 entities.
- **Order within each stratum:** `blake2b("<seed>:<entity_id>")`. Quotas use floor plus largest remainder.
- **Leakage prevention:**
  - Each S2/S3 target belongs to exactly one S1, re-checked at build time, and **inherits its S1's fold**.
  - Distractors are split 20/80, stratified by source × country.
  - So every record is in exactly one fold, and the fit and validation pools are disjoint. Verification found 0 fit targets inside the validation pool and 0 S1 in both folds.
- **Validation candidate pool:** all targets of the validation S1 plus their share of the distractors, giving 2,064,020 S2+S3 records (4.676 per S1; full train 4.677). It reproduces the *train* composition, not test's 5.75 per S1. The report lists this as a known limitation: validation may be optimistic on precision.
- The test files are never read by Step 1.

3.2 Validation statistics (measured)

	All train	Fit (80%)	Validation (20%)
S1 entities	2,206,821	1,765,457	441,364
Singletons	123,247	98,598	24,649 (5.585%)
True pairs	7,638,365	6,110,715	1,527,650
Pool S2 / S3	5,034,616 / 5,285,603	4,027,964 / 4,228,235	1,006,652 / 1,057,368
Validation targets / distractors	—	—	1,527,650 / 536,370

Country shares match the full data to within 0.01 percentage points (US 59.98%, India 40.02%).

3.3 Local scorer

- `src/ber/metrics.py` implements the official per-entity macro F0.5, including the singleton rules above. `src/score.py` is its command-line wrapper, with optional official-validator checks. `src/ber/ids_tsv.py` handles the TSV input/output format.
- A vectorised copy, `src/ber/matcher.py::entity_f05`, is used in the experiments. It is cross-checked against `src/score.py` on every validation prediction file; in Step 3 the two values were identical (0.962847).
- **Sanity scenarios** (`experiments/validation_sanity.py` → `experiments/validation_split/sanity_results.json`, log `experiments/validation_sanity_run.log`). For each, the validator passed and the scorer equalled both an exact recomputation and, where available, a closed-form value:
  - perfect = 1.000000
  - all-empty = 0.055847
  - keep-one = 0.696389
  - keep-half = 0.878121
  - add-distractor = 0.751544
  - rotate = 0.003145
- **Unit tests:** `src/tests/` has 60 `test_*` functions: blocking 10, ids_tsv 7, matching 18, metrics 20, validation_split 5. The notes record 32 tests passing at the end of Step 1.
  - A pass/fail record for the current full suite is **UNRESOLVED**; no log of a full-suite run was found.
  - `src/ber/normalize.py` refers to `src/tests/test_normalize.py`, which does **not** exist.
- The official validator passes on `val_ground_truth.tsv` against `val_as_test_dir` with `--check-ids`.

3.4 Artifacts

- `experiments/validation_split/assignment/s{1,2,3}_fold.npy`: fold per training-file row (0 = fit, 1 = validation).
- `experiments/validation_split/val_as_test_dir/test_source{1,2,3}.tsv` and `val_ground_truth.tsv`: validation laid out like the test directory.
- `manifest.json`, `verification.json`, `sanity_results.json`, `VALIDATION_SPLIT.md`.

4. Step 2: blocking and candidate generation

> **Key principle:** a TRUE MATCH must enter the candidate set, otherwise the matcher can never recover it. Candidate recall is therefore a hard ceiling on end-to-end recall. Every candidate is also a chance for a false merge, which matters a lot under F0.5.

4.1 Normalisation (`src/ber/normalize.py`, NORM_VERSION 2, identical to the EDA)

- General: NFKC, casefold, Latin diacritics stripped, punctuation turned into spaces, `&` → `and`.
- **Name keys:**
  - `name_raw`: the stripped raw name.
  - `name_n1`: the basic-normalised name.
  - `name_n2`: `name_n1` with runs of single letters joined and legal-form tokens removed (inc, llc, ltd, pvt, private, corp, sarl, sas, …, plus some Devanagari forms).
  - `name_n3`: the tokens of `name_n2`, sorted, so word order doesn't matter.
  - `name_tok1`: the first token of `name_n2`.
- **Address keys:**
  - `addr_n1`: the normalised comma-separated components.
  - `addr_n2`: the sorted set of address tokens, with street and state abbreviations expanded (Rd → road, NC → north carolina).
  - `addr_digits`: the sorted set of digit groups, with leading zeros removed.
  - `postal`: a 5- or 6-digit code.
  - `house`: the house number, with leading zeros stripped.
  - `country`: casefolded.
- Flags: name missing, address missing, non-Latin name, domain-like name, junk prefix, postal present, landmark.
- Retrieval tokens:
  - name: the words of `name_n2` plus character trigrams of the joined name
  - address: alphabetic tokens, digit groups and the house number

4.2 Selected configuration (`experiments/blocking/blocking_config.json`, SHA-256 `91a38976…0e5f5`)

Channel	What it does and why	Limit	Source / country handling
retrieval: nameaddr @10	TF-IDF cosine (binary term frequency, smooth IDF from the pool) over name+address tokens. It is the main recall channel: 93.97% recall alone on validation.	Top 10 per source; rarest query tokens scanned first, up to 20,000 postings	S2 and S3 searched separately (up to 20 per S1); only within the query's country
retrieval: addr @10	TF-IDF over address tokens only. It recovers pairs whose names differ (typos, trade names, scripts): 87.24% alone.	Top 10 per source; 20,000 postings	Per source; within country

exact: name_n1, name_n2, name_n3	Equality on normalised name variants (casing, legal form, word order)	Cap 100: any block with more than 100 pool records for a key is dropped	Joined per source. Country is part of a key only where listed; names and addresses already encode country.
exact: addr_n1, addr_n2	Equality on the normalised or reordered address		
exact: name_n2+country, addr_n2+country	Country-qualified variants		
exact: name_n1+addr_n1, name_n2+addr_n2, name_n3+addr_n2, name_n3+house	High-precision name+address combinations		
exact: name_tok1+house, name_tok1+addr_digits, postal+name_tok1, postal+house, house+addr_digits	Recover pairs with partial name agreement anchored on address numbers		

The candidate set is the **union** of all channels. Design rules, applied on the **fit fold only** with 100,000 fit S1 queries:

- **Postings budget:** the smallest budget within 0.1 pp of the largest budget's recall at 50. 20k was chosen over 100k and 500k.
- **Union growth:** keep adding while the marginal gain is at least 0.02 pp per extra candidate per S1, with the mean capped at 100 per S1. The result was `k_nameaddr=10, k_name=0, k_addr=10, exact=True`: fit recall 96.10% at 70.1 per S1.

4.3 Measured results

Metric	Validation (validation pool)	Stress (full training pool)	Test (no labels)
Candidate recall (union)	**98.214%** (S2 98.148%, S3 98.277%)	95.631%	**UNRESOLVED**
Matched S1 with all / at least one match retrieved	94.51% / 99.83%	87.22% / 99.54%	—
Candidate pairs	25,826,498	32,131,328	126,274,337
Candidates per S1: mean / median / p99 / max	58.5 / 39 / 208 / 390	72.8 / 42 / 250 / 437	72.9 / 42 / 255 / 533
Reduction ratio (vs full Cartesian product)	0.9999716	0.9999929	—
S1 with zero candidates	0	0	0

Recall of each channel alone on validation:

- nameaddr@10 93.97%; addr@10 87.24%; name@10 71.31% (not used).
- house+addr_digits 36.44% at cap 100 (59.78% with no cap).
- name_n3 52.50%.
- The 16 exact keys together (cap 100): 81.72% on the fit fold.

4.4 Failure cases and tradeoffs (measured)

Case (true pairs)	Validation union recall	Stress union recall
Target name non-Latin (110,516)	89.26%	80.75%
Target address missing (67,222)	90.02%	78.51%
House number differs (269,280)	95.26%	88.93%
Name other difference (723,947)	96.23%	91.03%
Domain-like target name (78,737)	98.08%	91.72%
Casing/punctuation, legal suffix, word order, address reorder	100.00%	99.47–100%

- In a sample of 40 validation misses the tags were: house number differs (23), other address difference (22), no shared name token (17), non-Latin / transliterated (14), typo-like (13), address missing on one side (12).
- Tradeoff: recall falls as the pool grows, because top-K retrieval competes with more records. The test pool is test-sized, so the test recall ceiling is probably closer to the full-pool figure than to validation. This is a hypothesis; test

recall is UNRESOLVED.

Files: `experiments/blocking/BLOCKING_REPORT.md`, `checkpoints/*.json`, `val_candidate_pairs.tsv` (339 MB), `experiments/blocking_experiment.py`, `src/ber/blocking.py`, `src/ber/features.py`, `src/generate_candidates.py`.

5. Step 3: the matcher

5.1 Features (50 per pair; `src/ber/match_features.py`)

Group	Features	Notes
Source	`is_s3`	Keeps S2 and S3 distinguishable
Exact-key equality	`eq_` + name_raw, name_n1, name_n2, name_n3, name_tok1, addr_n1, addr_n2, addr_digits, postal, house, country	0 if either side is empty
Presence indicators	`postal_both`, `house_both`, `digits_both`	Both sides have the field
Missing-value and quality flags (target)	`t_name_missing`, `t_addr_missing`, `t_name_nonlatin`, `t_name_domain`, `t_name_junk`, `t_postal`, `t_landmark`	Flags of the S2/S3 record
Query flags	`q_postal`, `q_landmark`	Flags of the S1 record
Name similarity (words + character trigrams)	`name_` + nq, nt, inter, jacc, cont_q, cont_t, cos, wjacc	Sizes, intersection, Jaccard, containment both ways, IDF cosine, IDF-weighted Jaccard
Address similarity	`addr_` + the same eight	Tokens: words, digit groups, house number
TF-IDF	`nameaddr_cos`	IDF = log((N+1)/(df+1)) + 1, computed from the pool being matched, so train, validation and test each use their own pool
Candidate rank (within S1)	`rank_src`, `gap_src`, `n_cand_src`, `rank_all`, `gap_all`, `n_cand_all`	Rank, gap to the best and group size of `nameaddr_cos` among the S1's candidates, per source and overall
Target context	`t_rank`, `t_gap`, `t_ncand`	Across **all S1 that retrieved the same target record**: this S1's rank (0 = best), the gap to the best S1's score, and the number of competing S1. Uses no labels. It depends on the true owner of each target being among the queries.

There is no phone, e-mail or website field in the data.

5.2 Model (`src/ber/gbdt.py`)

- A histogram GBDT written from scratch in numba (no GBDT library was available), with binary log-loss.
- Settings:
  - **300 trees, depth 6, learning rate 0.1**
  - L2 λ = 1.0, minimum child hessian 20
  - **64 quantile bins**, with edges taken from a 400k-row seeded sample
  - seed 0
- **Deterministic:** no row or feature subsampling.
- **Step 3 training data** (leakage-safe): an internal set **A** of 441,364 fit S1 with a validation-sized pool of their targets plus fit distractors.
  - A1 (353,091 S1) is used for fitting; A2 (88,273 S1) is used only for selection.
  - Rows: all 1,200,294 positives plus a deterministic 10% hash sample of negatives (1,942,608) with weight 10.
- **Selection** maximised the official macro F0.5 on A2, never on validation. The candidates were logistic regression and GBDT at 50/100/150/200/300 trees, each with and without the one-to-one rule. Chosen: GBDT, 300 trees, one-to-one ON, threshold 0.69 (A2 F0.5 0.96289).
- **Gain importance:** `t_rank` 76.5%, `addr_cont_t` 9.1%, `t_gap` 6.1%, `nameaddr_cos` 1.6%, and all others below 1%.

5.3 Decision rule (`submission_package/.../src/config/matcher_config.json`)

1. Predict a match if probability ≥ **0.69**.
2. **One-S1-per-target:** if several S1 claim the same S2/S3 record, only the highest-probability S1 keeps it. This is justified because no training target belongs to more than one S1.

3. No other post-processing.

5.4 GBDT vs logistic regression (validation, held out)

Metric	GBDT (chosen)	Logistic regression (t = 0.71, 1:1)
Official macro F0.5	**0.96285**	0.94458
Pair precision / recall	98.22% / 94.08%	97.50% / 90.79%
Singleton F0.5	91.93%	86.20%

The validation-optimal threshold would have been 0.70 (0.96291), so the A2-chosen 0.69 transfers.

6. Validation results (Step 3 model, held-out validation, 441,364 S1)

Metric	Value	Meaning
Candidate recall	98.214%	True pairs present in the candidates (the ceiling)
Pair precision	98.22%	Share of predicted pairs that are true (1,437,174 of 1,463,221)
Pair recall (end-to-end)	94.08%	Share of all true pairs predicted, including pairs never blocked
Pair recall among blocked	95.79%	Recall over true pairs that were candidates
Pair F0.5 / F1	97.36% / 96.10%	Pair-level (not the official metric)
Official macro F0.5	**0.96285**	Competition metric, per S1 then averaged. Cross-checked with `src/score.py` and the official validator.
Entity precision / recall	98.01% / 94.04%	Mean per-S1 precision (over S1 with a prediction) and recall (over S1 with true matches)
Singleton F0.5	91.93%	Share of singleton S1 left correctly empty
Singleton false-merge rate	8.07%	Singletons that received any prediction (each scores 0)
F0.5 on S1 with matches	96.54%	—
S1 predicted exactly right	78.93%	The predicted set equals the true set
By country	US 0.9712 / India 0.9503	France cannot be validated (no labels)
By source (pair precision / recall)	S2 98.11% / 93.78%; S3 98.32% / 94.36%	—

7. Error analysis (observed)

Counts on validation (Step 3):

- TP 1,437,174.
- FP 26,047, of which 2,110 are on singleton S1.
- FN blocked but rejected by the matcher: 63,199.
- FN never in the candidates: 27,277.

Source: `experiments/matching/MATCHING_REPORT.md` §F and `errors.json`.

Share of pairs with property	TP	FP	FN (blocked)
Same name, different address (name_cos ≥ 0.8 & addr_cos < 0.3)	2.92%	16.43%	27.08%
Same address, different name (addr_cos ≥ 0.8 & name_cos < 0.3)	6.36%	11.16%	4.54%
House numbers both present but different	15.75%	49.56%	47.76%
Target address empty	2.75%	18.07%	33.22%
Target name non-Latin	6.07%	16.82%	18.11%
Target name domain-like	5.35%	1.08%	0.60%
Target name junk prefix (noisy prefixes like `--`, `#`)	2.52%	1.22%	0.97%
Landmark on either side	5.33%	6.47%	6.65%
Query is a singleton	0.00%	8.10%	0.00%

- **Legal-suffix differences:** almost all of the top-confidence false positives (p ≈ 1.000) are records that differ only in legal form or by an extra token. Examples from the report: "Corner Tattoo" vs "Corner Tattoo Co"; "Delta Capital LLC" vs "Delta Capital Corp.". The labels say these are *different* entities.
- **Singleton errors:** 8.07% of singletons are falsely merged, mostly with near-identical records (legal-suffix or token-subset variants).
- **False negatives:** native-script names without transliteration, empty target addresses, changed house numbers, typo-heavy names, and token subsets such as "Psy Management Private Limited" vs "Psy Management Limited Services".
- **Multiple true matches:** blocking recall on S1 with 2 or more true matches is 98.22% (validation). Matched S1 with *all* true matches retrieved: 94.51%.
- No cross-country errors: 0% in every group.

8. Test-scale and stress evaluation

8.1 Why and how

- **Why:** the test pool is about 5× the validation pool and has a different S1-to-pool ratio. The model's target-context features depend on which S1 compete for each target.
- **Unlabelled test check** (`experiments/test_scale/TEST_SCALE_REPORT.md`): the Step 3 model was run on test, *counts only*.
  - 126,274,337 candidates.
  - Test S1 per pool record is 0.1738, against 0.2138 on validation.
  - Target-context feature drift is **negligible**: PSI `t_rank` 0.0297, `t_ncand` 0.0709, `t_gap` 0.0719. `n_cand_all` shows moderate drift (0.22).
- **Labelled stress evaluation** (`experiments/TEST_SCALE_REPORT.md`, `experiments/stress_matcher_eval.py`): the *same* 441,364 validation S1 matched against the **full training pool** (10,320,219 records).
  - The 8.26M added fit-fold records are true non-matches, so the labels stay complete.
  - This setup has 0.043 S1 per pool record, far more "orphaned" than test's 0.174.

8.2 What happened (measured)

Metric	A: normal validation	B: stress
Official macro F0.5	0.962847	**0.840024**
Pair precision / recall	98.22% / 94.08%	82.45% / 93.14%
Singleton F0.5 / false-merge rate	91.93% / 8.07%	42.53% / 57.47%
Predicted pairs / FP	1,463,221 / 26,047	1,725,775 / 302,910
Candidate recall; candidates per S1	98.214%; 58.5	95.631%; 72.8
Share of pairs with `t_rank` = 0	7.73%	20.80%

- **Orphaned targets:** 272,558 of the 302,910 false positives (90.0%) are on fit-fold records, whose owner S1 is not queried in B. That lets some validation S1 become the "best S1" for the target (`t_rank` = 0) by default.
  - The later diagnostic (§9) split these into **183,185 on fit-fold true targets** (genuine orphans) and **89,373 on fit-fold distractors**.
- Restricting the same predictions to validation-pool records gives **0.95603**, a decomposition rather than a deployable configuration (`experiments/test_scale/stress/orphan_decomposition.json`).
- Target-context features shifted **majorly** in B (PSI `t_gap` 1.18, `t_ncand` 0.92, `t_rank` 0.56). On real test the same features shifted only negligibly.
- All 28 Step 1–3 files had the same hashes before and after.

8.3 Conclusions

Justified:

- The matcher is stable when each target's owner is among the queries.
- It is not robust when owners are missing: it makes confident false merges, especially on legal-form near-duplicates.
- The GBDT relies heavily on `t_rank`.

NOT justified:

- That test F0.5 will be 0.840. B is a pessimistic environment, not a replica of test.
- Anything about France. There are no labelled France records.

9. `t_rank` diagnostic COMPLETE

Script `experiments/trank_diagnostic.py`; outputs in `experiments/diagnostics/trank/`: `TRANK_DIAGNOSTIC_REPORT.md`, `evaluate.json`, `breakdown.json`, `select.json`, `train.json`, `threshold_curves.json`, `run.log`. It ran from 2026-09-26 01:25 to 11:47 with exit code 0; the machine slept for part of that time. All 34 protected production files had the same hashes before and after. The model A probabilities reproduced the saved production values exactly (max difference 0) on V and S.

9.1 Design

- **Environments** (validation S1 and labels only):
  - **V:** normal validation.
  - **S:** the stress run above.
  - **T (new):** the validation pool plus 475,723 random fit-fold records (351,950 fit targets, i.e. orphans, and 123,773 distractors). This makes 5.754 pool records per S1, which gives 0.1738 S1 per pool record, **equal to test**. Blocking was rerun with the unchanged configuration: 26,368,048 candidates, recall 97.99%.
- **Variants**, all trained exactly like Step 3 (A1 rows, 300 trees) with the threshold chosen on A2 and the one-per-target rule:
  - **A:** the current model (the saved Step 3 model).
  - **B:** without `t_rank`.
  - **B2:** without `t_rank`, `t_gap` and `t_ncand`.
  - **C:** `t_rank` capped at 1. Monotonic transforms such as a log have no effect on trees.
  - **D:** `t_rank` / max(`t_ncand` − 1, 1), a percentile.

9.2 Results (measured)

Env	Var	A2 threshold	Macro F0.5	Precision	Recall	Singleton F0.5	FP	FN	FP on orphans
V	A	0.69	**0.96285**	98.22%	94.08%	91.93%	26,047	90,476	0
V	B	0.68	0.96281	98.20%	94.13%	91.77%	26,361	89,660	0
V	B2	0.71	0.95527	98.67%	90.90%	93.64%	18,681	139,035	0
V	C	0.70	0.96292	98.28%	93.94%	92.38%	25,107	92,555	0
V	D	0.70	0.96295	98.32%	93.87%	92.45%	24,534	93,662	0
S	A	0.69	**0.84002**	82.45%	93.14%	42.53%	302,910	104,785	183,185
S	B	0.68	0.83753	82.16%	93.19%	41.36%	309,067	103,997	186,819
S	B2	0.71	0.90154	93.22%	89.03%	69.32%	98,948	167,605	27,834
S	C	0.70	0.84303	82.73%	93.04%	44.77%	296,644	106,315	181,118
S	D	0.70	0.84347	82.87%	93.02%	44.12%	293,643	106,693	178,920
T	A	0.69	**0.95432**	97.22%	94.12%	87.79%	41,186	89,791	9,896
T	B	0.68	0.95420	97.19%	94.17%	87.65%	41,585	89,080	10,014
T	B2	0.71	0.95125	98.28%	90.79%	91.73%	24,308	140,670	2,561
T	C	0.70	0.95471	97.30%	93.98%	88.52%	39,853	91,980	9,734
T	D	0.70	0.95482	97.35%	93.92%	88.41%	39,033	92,838	9,471

FN includes pairs never blocked: 27,277 in V, 66,750 in S and 30,686 in T.

Top gain features by variant:

- B: `t_gap` 81.4%.
- B2: `addr_jacc` 68.4%, `nameaddr_cos` 11.3%.
- C: `t_rank_cap1` 73.9%.
- D: `t_gap` 50.0%, `t_rank_pct` 31.5%.

Macro F0.5 against the threshold (one-per-target rule ON; `threshold_curves.json`; diagnostic only, not a selection):

Env / variant	0.69	0.72	0.75	0.78	0.81	0.85	0.90	0.95

V / A	0.9628	0.9629	0.9626	0.9619	0.9606	0.9579	0.9515	0.9374
T / A	0.9543	0.9552	0.9557	0.9559	0.9555	0.9540	0.9491	0.9366
S / A	0.8400	0.8482	0.8567	0.8655	0.8749	0.8872	0.9019	0.9117
V / B2	0.9556	0.9550	0.9540	0.9526	0.9507	0.9469	0.9391	0.9219
T / B2	0.9513	0.9512	0.9506	0.9497	0.9480	0.9448	0.9374	0.9205
S / B2	0.8983	0.9031	0.9073	0.9110	0.9140	0.9163	0.9155	0.9034

Current model A in environment S, by category (`breakdown.json`):

- 300,803 of the 302,910 FP (99%) have `t_rank` = 0.
- Precision in that bucket: 82.53% in S, against 97.24% in T.
- Precision with an empty target address: 41.62% in S / 84.11% in T.
- Precision on same-name, different-address pairs: 47.51% in S / 86.87% in T.
- Precision with different house numbers: 68.76% in S / 92.73% in T.
- Country (macro F0.5 in S): US 0.8623, India 0.8066.
- S2 and S3 behave the same (about 82% precision each in S).

9.3 What the diagnostic supports

All of the following are **measured**, on single deterministic runs with no repeated seeds or confidence intervals:

- `t_rank` **alone is not the cause.** B, C and D stay within ±0.004 of A in every environment. When `t_rank` is removed, the model moves its reliance to `t_gap`.
- **The target-context family as a whole is orphan-sensitive.** Removing all three (B2) raises S by 0.0615 but lowers V by 0.0076 and T by 0.0031. At the test-like ratio these features still add value.
- **In S the damage is mostly overconfident probabilities.** Model A at threshold 0.95 reaches 0.9117 in S, close to B2's best of about 0.916. The best threshold rises with orphan density: about 0.71 in V, 0.77 in T and 0.96 in S.
- **At the test-like ratio, the orphan effect costs about 0.009** (V 0.9628 → T 0.9543). The rest of the gap to the leaderboard score is **UNRESOLVED**.

Caveats:

- T assumes the extra test records resemble random fit-fold records. Their true nature is **UNRESOLVED**.
- No statistical significance testing was done. Differences below about 0.001 should be treated as noise-level. **This is a judgement, not a measured value.**

10. Final training and inference

Sources: `experiments/final/FINAL_REPORT.md`, `final_run.log`, `timings.json` and `*.json`; script `experiments/final_pipeline.py`. The run started 2026-09-25 18:22, was resumed at 23:14 and finished 2026-09-26 00:03 with exit code 0.

Item	Value (measured)
Training data	All 2,206,821 training S1 blocked against the full training pool with the Step 2 config: 160,550,236 candidate pairs, candidate recall 95.646%.
Training rows	22,629,394: all 7,305,755 positives plus 15,323,639 negatives (10% hash sample, weight 10).
Model	Step 3 GBDT parameters (300 trees, depth 6, learning rate 0.1, λ 1, minimum child weight 20, seed 0). Threshold 0.69, one-per-target ON. Model SHA-256 `1bac9115…66755`; file `experiments/final/model_final.npz`.
Integrity checks	The Step 3 model, selection and blocking config matched their recorded hashes before training.
Test candidates	126,274,337 pairs for 1,732,544 S1, reused from the test-scale run. Per S1: mean 72.9, median 42, p99 255, max 533; 0 S1 without candidates.
Test inference	126,274,337 pairs scored in 766.5 s.
Predictions	5,523,440 before the rule; 19,970 removed by the one-to-one rule; **5,503,470 final** (S2 2,652,224, S3 2,851,246). 3.177 per S1.
No-match S1	**110,945 (6.40%)**. By country: US 6.13%, India 7.20%, France 4.62%. Predictions per S1: US 3.207, India 3.042, France 3.518.
Step 3 model on test (diagnostic, for comparison)	6,254,297 predictions, 3.68% zero-prediction. Retraining on all data made the final model more conservative on test.
Runtime	Phase seconds: • train_candidates 2,614.8; train_idf 17.3; train_cos 786.7; train_tc 133.3; train_rows 514.8; train_model 1,076.8 • test_inputs 2.9; test_infer 767.2; decide 4.5; write 109.5; validate 463.0 • about 1.8 h in total, *excluding* the test featurisation and candidate generation reused from the test-scale run (featurise 316.8, candidates 3,331.7, cos 991.9, target context 124.9)
Memory	Peak working set 7.99 GB (it includes memory-mapped pages); private memory about 1.3–2.1 GB.
Disk	`experiments/final/` holds about 8.5 GB of intermediates (training feature matrix `train_X.f32` 4.5 GB, candidate and context arrays), plus the 1.65 GB `candidate_pairs.tsv`; 10.1 GB in total. • The package README says **about 22 GB** of work directory is needed. • The `--work-dir` help text in `run_pipeline.py` says **about 40 GB**. • The two figures are **inconsistent**; **UNRESOLVED**.

11. Submission

Item	Value
File uploaded to the portal	`output/matching_results.tsv`: 93,375,709 bytes, SHA-256 `84ede6d4f2e44ab4469ea7798a95fbd8b3c49fc0f3f1561d47064c1caaa0e807`.
Format	Header `source1_entity_id<TAB>matched_entity_ids`; 1,732,544 rows (110,945 empty, 1,621,599 non-empty); comma-separated IDs with no spaces.
Candidate file	`output/candidate_pairs.tsv`: 1,649,916,409 bytes, SHA-256 `1842a3df…4835ea`, identical to `experiments/final/candidate_pairs.tsv`. 1,732,544 rows, 0 empty.
Official validator	**PASS**, exit 0, with `--matching`, `--candidate` and `--check-ids` (9,969,589 valid S2/S3 IDs). No matched ID falls outside the candidates. It was rerun on 2026-09-26 just before packaging.
Zip	`camelcase_submission.zip` at the project root: 728,869,951 bytes, 17 entries, CRC test OK. The output files inside match the original hashes. • `output/matching_results.tsv`, `output/candidate_pairs.tsv` • `Documentation_template.md` (filled) • `code/business_entity_resolution/` with `README.md`, `requirements.txt`, `src/run_pipeline.py`, `src/ber/` (9 modules) and `src/config/` (`blocking_config.json`, `matcher_config.json`) • Caches were removed and there are no local paths or secrets.
Environment (pinned)	Python 3.11.9, numpy 2.4.6, numba 0.65.1, llvmlite 0.47.0, scipy 1.17.1 (`submission_package/.../requirements.txt`).
Local validation score	**0.96285**: held-out training labels, Step 3 model.
Competition leaderboard score	**0.924**: public subset of test, final model. Team-reported and not stored in any file. *These are different populations and different models; do not compare them as like for like.*
Zip uploaded as the final package?	**UNRESOLVED**. Only the upload of `matching_results.tsv` is recorded, in the session.

12. What we currently know

A. Experimentally confirmed (measured)

1. The GBDT beats logistic regression on validation (0.96285 vs 0.94458). 300 trees beat 50–200, and the one-to-one rule helps slightly (A2).
2. Blocking recall is 98.214% on validation. It falls to 95.631% against a test-sized pool; the ceiling drops as the pool grows.
3. Legal-form, casing, word-order and address-reorder variants are 100% recalled by blocking. Native-script names (89.26%) and empty target addresses (90.02%) are the weakest blocking cases.
4. The stress environment drops F0.5 to 0.840, and 90% of its false positives are on fit-fold records (183,185 orphaned targets and 89,373 fit distractors).
5. Restricted to validation-pool records, the stress predictions score 0.956.
6. Removing, capping or taking the percentile of `t_rank` changes F0.5 by no more than about 0.004 in V, S or T.
7. Removing all target-context features helps S (+0.062) and hurts V (−0.0076) and T (−0.0031).
8. At the test S1-to-pool ratio (T) the current model scores 0.9543.
9. The best threshold rises with orphan density (about 0.71 in V, 0.77 in T, 0.96 in S). In T, threshold 0.75–0.78 gives 0.9557–0.9559 against 0.9543 at 0.69.
10. Target-context feature drift between validation and real test is negligible (PSI ≤ 0.072).
11. Final model on test: 5,503,470 predictions and 6.40% zero-prediction S1. The validator passes.
12. India is consistently weaker than the US: validation 0.9503 vs 0.9712, and every environment agrees.

B. Suspected but NOT proven (hypotheses)

1. France (15% of test, no labels) causes a large part of the remaining gap of about 0.03. **If the non-France 85% scored like T (0.954), France would need about 0.75 to give 0.924.**
2. France is over-merged. Final-model test statistics: France 3.518 predictions per S1 and 4.62% zero-prediction, against US/India 3.04–3.21 and 6.1–7.2%. For reference, the training singleton rate is 5.58%. This is suggestive only.
3. The extra test pool records (5.75 vs 4.68 per S1) behave like orphans or distractors, lowering precision.
4. The test candidate recall ceiling is nearer the full-pool figure (about 95.6%) than validation's 98.2%.
5. Validation is optimistic on precision, because it reproduces the train pool composition (VALIDATION_SPLIT.md).
6. Near-duplicates that differ only in legal form are intentionally distinct entities in the data generator, so name-similarity features alone cannot separate them.

C. Not tested yet

1. Leave-one-country-out (train US, evaluate India, and the reverse) as a proxy for unseen-country robustness.
2. Orphan-aware training, i.e. training with fit-fold orphans added to the internal training pool.
3. Any threshold other than 0.69 on the leaderboard. No alternative model or threshold has been submitted.
4. GBDT hyperparameters other than tree count: depth, learning rate, λ, minimum child weight and bins were never tuned.
5. Transliteration or script-aware handling of native-script names, in blocking or features.
6. Special handling of empty-address targets.
7. The end-to-end F0.5 effect of larger retrieval K, or an extra name channel. Only candidate recall was measured.
8. Pretrained or embedding models (allowed if MIT/Apache and ≤ 8B; none used).
9. A full-scale run of the packaged `run_pipeline.py`.
10. Statistical significance or seed variance for any comparison.

13. Improvement opportunities (not ranked)

Each direction below is backed only by existing evidence. None has been run. Unless stated otherwise, compare the official macro F0.5 on V and T, and check S as a robustness guard.

Direction	Hypothesis	Evidence for	Evidence against	Risk	Experiment, metric and success criterion
Unseen-country (France) robustness	France accounts for much of the remaining gap of about 0.03	T (0.954) is well above the leaderboard (0.924); France has more predictions per S1 and fewer empty predictions	No labels; the gap could come from other test differences	Changes aimed at France can't be validated directly	Leave-one-country-out training (US→India, India→US) with the V/T setup. Measure the drop from in-country to cross-country F0.5, and test whether a candidate fix reduces it without lowering in-country V.
Orphan-aware training / calibration	Training with realistic competition makes the target-context features calibrated	S damage is mostly calibration (A at t = 0.95 reaches 0.912); the best threshold shifts with orphan density	At the test ratio the orphan cost is only about 0.009	May lower V	Rebuild set A with fit-fold orphans at the test ratio and retrain (Step 3 protocol). Success: T and S improve, and V drops by no more than about 0.001.
Threshold hedging	A slightly higher threshold is more robust to test shift	T: 0.75–0.78 gives +0.0014–0.0016; S: +0.017–0.026; V costs only 0.0002–0.0009	If test behaves like V, there is a small loss; unverified on the leaderboard	Low	Compare thresholds 0.69–0.81 on V/T/S using saved probabilities. Only a leaderboard submission can confirm; one controlled upload.
Native-script names	Transliteration or script-aware tokens would recover missed pairs	Blocking recall 89.26% (validation) and 80.75% (stress); 18.11% of FN and 16.82% of FP involve non-Latin targets	About 5% of target names; must be learned from the training data only (no external transliteration data)	Must avoid external resources	Measure blocking recall on the non-Latin category first, then end-to-end F0.5 on V/T.
Empty target addresses	Dedicated handling reduces FN and FP	Recall 90.02%; FP share 18.07% and FN share 33.22%; S precision 41.6%	Small share of pairs (about 3.3% of records)	More false merges	A category-specific feature or threshold, tuned on A2. Measure category precision and recall and the macro F0.5 on V/T.
House-number conflicts and legal-form near-duplicates	Better conflict features reduce confident FP	House-number differences appear in 49.56% of FP and 47.76% of FN; the top-confidence FPs are legal-suffix variants	The same patterns occur in TP (15.75% house differ)	Recall loss	Feature ablation or additions under the Step 3 protocol. Measure FP/FN in these categories and the macro F0.5.
Blocking recall at test scale	A larger K or an extra channel raises the ceiling	Recall drops from 98.21% to 95.63% with a larger pool; name@50 and nameaddr@50 are higher (96.30%)	More candidates bring more false-merge opportunities; the current design rule limited growth	Runtime, disk and precision	Candidate recall in S/T, then end-to-end F0.5 on V/T/S. Success: F0.5 rises, not just recall.

14. Experiment history (chronological)

When (from file times)	Change / experiment	Why	Result	Kept?	Files
09-25 00:17–08:10	Competition notes, streaming EDA	Understand rules and data	Measured facts (§1.5); PIN/ZIP blocking not viable (postal codes in < 3% of records)	Kept	`docs/COMPETITION_NOTES.md`, `experiments/EDA_REPORT.md`
09-25 08:20–08:31	Step 1 split, scorer, sanity checks	Leakage-safe local evaluation	441,364 validation S1; sanity OK	Kept	`experiments/validation_split/`
09-25 08:47–09:22	Blocking on smoke data	Debugging	Pipeline works	Smoke only	`experiments/_smoke/blocking*`
09-25 12:17–12:59	Blocking design (fit fold) and validation	Choose channels and budgets	Budget 20k chosen over 100k/500k; name channel K = 0 (not used); nameaddr and addr K = 10; exact cap 100. Validation recall 98.214%.	Kept (the alternatives were rejected by the pre-stated rules)	`experiments/blocking/`
09-25 13:20	Blocking stress (full pool)	Test-scale recall	95.631%	Info	`checkpoints/stress.json`
09-25 14:11–14:36	Step 3 matcher	Model selection on A2	GBDT@300 + 1:1 + 0.69 chosen; logistic regression and GBDT@50/100/150/200 rejected; validation 0.96285	Kept	`experiments/matching/`
09-25 16:01–17:46	Unlabelled test-scale check	Test compatibility and drift	t-feature PSI negligible	Info	`experiments/test_scale/`
09-25 17:52–18:12	Labelled stress evaluation	Robustness at test scale	0.840; orphan decomposition 0.956	Info (no change)	`experiments/TEST_SCALE_REPORT.md`
09-25 18:22 – 09-26 00:03	Final retrain on all training data and test inference	Produce the submission	5,503,470 predictions; validator PASS	Kept (submitted)	`experiments/final/`, `output/`
09-26 00:10–01:01	Package, documentation, zip	Final package	Zip verified	Kept	`submission_package/`, `camelcase_submission.zip`
09-26 (session)	Leaderboard upload	Score	Public 0.924 (team-reported)	—	not in files
09-26 01:25–11:47	`t_rank` diagnostic (B, B2, C, D; environment T)	Explain the gap	§9	Nothing adopted; no selection made	`experiments/diagnostics/trank/`

Please don't repeat these: `t_rank` removal, capping and percentile (no effect); full removal of the target context (V/T get worse); tree counts of 300 or fewer and logistic regression (already compared); postings budgets of 100k/500k (no recall gain).

15. Reproduction guide

> **Do not run** `experiments/final_pipeline.py` with its defaults: `--submission-path` defaults to `output/matching_results.tsv` and `--out-dir` to `experiments/final`, so it would **overwrite the submitted file and the final model**. Always pass new output locations. Commands marked **[EXPENSIVE]** take tens of minutes to hours and need GB of RAM and disk. None of these commands were run to write this document.

```
# Environment: Python 3.11.9 + numpy 2.4.6, numba 0.65.1, llvmlite 0.47.0, scipy 1.17.1 (no pandas / sklearn
needed).

# Unit tests
python -m unittest discover -s src/tests

# Step 1 — rebuild / verify the validation split (reads training data only)
python src/build_validation_split.py --verify-only        # checks the saved split
python src/build_validation_split.py                      # rebuilds (seed 20260925)

# Score a validation prediction file with the official metric
python src/score.py -p <predictions.tsv> -g experiments/validation_split/val_ground_truth.tsv \
       -t experiments/validation_split/val_as_test_dir [--check-ids]

# Official format validator
python utils/validate_submission.py --matching <matching.tsv> --candidate <candidates.tsv> \
       --test-dir dataset/test --check-ids                # --check-ids loads ~10M IDs (a few GB RAM)

# Step 2 — blocking experiment [EXPENSIVE]  (phases: design, evaluate, stress, report, ...)
python experiments/blocking_experiment.py --phase <phase>
python src/generate_candidates.py --test-dir <dir> --config experiments/blocking/blocking_config.json \
       --features-dir <dir> --out <candidate_pairs.tsv>

# Step 3 — matcher experiment [EXPENSIVE; X_A / X_val are ~5 GB each]
python experiments/matching_experiment.py --phase all --out-dir <new dir>

# Unlabelled test-scale check [EXPENSIVE, ~1.7 h]
python experiments/test_scale_check.py --out-dir <new dir>

# Labelled stress evaluation [EXPENSIVE]
python experiments/stress_matcher_eval.py --out-dir <new dir> --report-path <new file>

# t_rank diagnostic [EXPENSIVE, ~1.5 h active compute, ~15 GB disk]
python experiments/trank_diagnostic.py --out-dir <new dir>

# Final pipeline (experiment version) [EXPENSIVE, ~1.8 h with reused test candidates; more from scratch]
python experiments/final_pipeline.py --out-dir <new dir> --submission-path <new file>

# Packaged, self-contained pipeline [EXPENSIVE; README states ~3.2 h; only smoke-tested so far]
cd submission_package/code/business_entity_resolution
python src/run_pipeline.py --train-dir <dataset>/train --test-dir <dataset>/test --work-dir work --output-dir
output
```

Small copies of the data and every stage exist under `experiments/_smoke/`. Use them to test code changes in seconds before running a full-scale job.

16. File and directory map

Path	Purpose
`dataset/train/`, `dataset/test/`	Official data (TSV). Never modify.
`docs/problem_statement.pdf`, `README.md`	Official problem statement; the root README is a markdown copy of it.
`docs/COMPETITION_NOTES.md`	Rules, spec, measured data facts, project status (§0). Start here.
`Documentation_template.md` (root)	Official unfilled template. The filled copy is in `submission_package/`.
`utils/validate_submission.py`	Official validator (stdlib only). Do not modify.
`src/ber/`	Library: • `normalize.py`: keys, tokens, flags • `features.py`: feature stores • `blocking.py`: exact joins and TF-IDF retriever • `blocking_eval.py` • `match_features.py`: 50 features • `gbdt.py` • `matcher.py`: decision rule, vectorised F0.5, logistic regression, TSV writer • `metrics.py`: official metric • `ids_tsv.py` • `validation_split.py`
`src/build_validation_split.py`, `src/score.py`, `src/generate_candidates.py`, `src/tests/`	Command-line entry points and unit tests.
`experiments/eda.py`, `EDA_REPORT.md`, `eda_results.json`, `_eda_cache/`	EDA and the ground-truth cache used by later stages.
`experiments/validation_split/`	Step 1 artifacts (§3.4).
`experiments/blocking/`	Step 2 config, report, checkpoints, validation candidates, feature stores (`features/`).
`experiments/matching/`	Step 3 model (`model_gbdt.npz`), `selection.json`, feature matrices, predictions, `errors.json`, report.
`experiments/test_scale/` (+ `stress/`)	Unlabelled test check and labelled stress run artifacts.
`experiments/TEST_SCALE_REPORT.md`	Labelled stress report. Note: `experiments/test_scale/TEST_SCALE_REPORT.md` is a different report (the unlabelled test check).
`experiments/final/`, `experiments/final_pipeline.py`	Final model, run log, report, final candidates; `packaging/` holds the package smoke test.
`experiments/diagnostics/trank/`, `experiments/trank_diagnostic.py`	`t_rank` diagnostic (about 15 GB of intermediates plus report and JSON).
`experiments/_smoke/`	Small-scale copies of all stages for fast testing.
`experiments/*.log`	Run logs of each stage.
`output/`	**Submitted** `matching_results.tsv` and `candidate_pairs.tsv`. Do not overwrite.
`submission_package/`, `camelcase_submission.zip`	Final package source and the built zip. Do not overwrite.
`docs/TEAM_CONTEXT.html` → `TEAM_CONTEXT.pdf`	This document (HTML source and PDF).

17. What a new team member should do first

1. **Read this PDF**, especially §0, §9 and §12.
2. **Read the key reports:**
   - `docs/COMPETITION_NOTES.md`
   - `experiments/validation_split/VALIDATION_SPLIT.md`
   - `experiments/blocking/BLOCKING_REPORT.md`
   - `experiments/matching/MATCHING_REPORT.md`
   - `experiments/TEST_SCALE_REPORT.md`
   - `experiments/final/FINAL_REPORT.md`
   - `experiments/diagnostics/trank/TRANK_DIAGNOSTIC_REPORT.md`
3. **Understand the baseline:** the Step 3 GBDT scores 0.96285 on local validation. The final model trained on all data is what was submitted, and it scored 0.924 on the public leaderboard (team-reported).
4. **Understand the known failure modes:** orphan-driven false merges on target-context features, legal-form near-duplicates, house-number conflicts, empty addresses, native-script names, and unlabelled France.
5. **Check the latest diagnostic results** (§9). The environments V, T and S are the agreed yardstick; T matches the test S1-to-pool ratio.
6. **Propose one controlled experiment at a time** (§13). Change one thing, keep the Step 3 protocol (fit on A1, choose on A2, report V/T/S), smoke-test in `experiments/_smoke/`, and write to a new output directory.
7. **Never overwrite the current submission or model before comparing results:** `output/`, `experiments/final/`, `experiments/matching/`, `experiments/blocking/blocking_config.json`, `submission_package/` and the zip. Hash them before and after, as the existing scripts do.

Unresolved items (collected)

- The leaderboard score (0.924) is only team-reported; it is not stored in any project file. The public/private split size, the team's rank, and whether the zip has been submitted are also unknown.
- Test candidate recall and France performance: no labels exist.
- The cause of the remaining gap of about 0.03 between T and the leaderboard.
- The nature of the extra test pool records (5.75 vs 4.68 per S1).
- Official F0.5 conventions when the true set is non-empty and the prediction is empty or has no overlap. Locally both are scored 0.
- Deadlines, rate limits, and compute and hardware limits.
- No full-scale run of the packaged pipeline, and the work-dir estimates disagree (README about 22 GB vs help text about 40 GB).
- No recorded pass/fail result for the current unit-test suite. `test_normalize.py` is referenced in the code but missing.